Master Recherche $(MS)^2SC$ 2025–2026

MG3416–Advanced Structural Dynamics and Acoustics

Tutorial class #2

Exercice 1: Acoustic cavity with dissipative wall

We consider an inviscid compressible barotropic fluid contained in a parallelepipedic rigid cavity, with its upper wall covered by a thin layer of viscoelastic material, in order to absorb some part of the acoustic energy of the fluid. The purpose of the absorber is for example to decrease the level of noise in aircrafts or cars submitted to external excitations. The boundary $\partial\Omega_{\mathrm{F}}$ of the domain $\Omega_{\mathrm{F}} =]0,a[\times]0,b[\times]0,c[$ occupied by this fluid is $\partial\Omega_{\mathrm{F}} = \Gamma_R \cup \Gamma_Z$, where Γ_R is constituted by the five rigid walls of the cavity, and $\Gamma_Z =]0,a[\times]0,b[\times\{z=c\}$ is the upper absorbing wall (see Fig. 1). The fluid pressure field is denoted by p, the fluid velocity field by $\boldsymbol{v}_{\mathrm{F}}$, its mass density by ϱ_{F} and the speed of sound by c_{F} (the fluid is assumed homogeneous). The absorbing boundary is such that the pressure field reads:

$$\partial_t p(\boldsymbol{x},t) = \varrho_{\mathrm{F}}\,(\kappa + \delta\partial_t)\,\boldsymbol{v}_{\mathrm{F}}(\boldsymbol{x},t)\cdot\boldsymbol{n}\,,\quad \boldsymbol{x}\in\Gamma_Z\,, t\in\mathbb{R}\,,$$

where $\boldsymbol{n}$ is the unit outer normal to $\partial\Omega_{\mathrm{F}}$, and κ, δ are strictly positive constants. At last the (harmonic) acoustic source inside the cavity is denoted by $\boldsymbol{f}$.

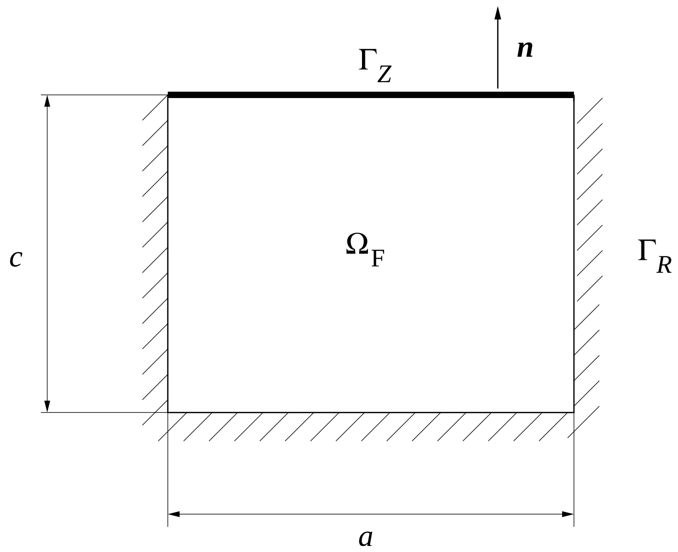

Figure 1: Rigid cavity with absorbing wall filled with an acoustic fluid.

Q1. What are the dimensions of κ and δ? What do they correspond to?

Q2. Write all equations satisfied by the fluid pressure and velocity fields in Ω_{F}.

Q3. Write the associated boundary-value problem for the fluid pressure in the frequency domain. The specific impedance of the absorbing boundary will be denoted by $\mathcal{Z}(\omega)$ such that $\mathrm{i}\omega\mathcal{Z}(\omega) = \kappa + \mathrm{i}\omega\delta$.

Q4. Write the weak formulation of the boundary-value problem of question 3.

Q5. Compute the eigenmodes and eigenfrequencies of the cavity with rigid walls by the method of separation of variables.

Q6. Apply the Galerkin method to this formulation, using (justify this choice) the eigenmodes computed in question 5.

Q7. What is the added effect induced by the viscoelastic layer for the acoustic cavity? Comment.

Q8. To go further (optional) *Compute the eigenmodes and eigenfrequencies of the cavity with the absorbing coating. How is the Galerkin formulation modified by using these modes?*

Exercice 2: Plane wave transmission at the interface between air and water

We consider the reflection and transmission of acoustic waves at the interface between two acoustic media at rest, namely air and water. Their interface Σ is an horizontal plane with unit normal $\boldsymbol{n}$ and both media are labelled by subscripts 1 and 2, respectively. The sound speeds and material densities are denoted by c_j and ϱ_j, respectively, $j = 1, 2$. The interface is impinged by an incident acoustic wave (circular frequency ω and wave vector $\boldsymbol{k}_\mathrm{I}$) $\boldsymbol{v}_\mathrm{I} = c_1 \hat{\boldsymbol{k}}_\mathrm{I}\,\mathrm{e}^{\mathrm{i}(\omega t - \boldsymbol{k}_\mathrm{I}\cdot\boldsymbol{x})}$ from the air in the direction $\hat{\boldsymbol{k}}_\mathrm{I} = \frac{\boldsymbol{k}_\mathrm{I}}{|\boldsymbol{k}_\mathrm{I}|}$ and it is reflected/transmitted at that interface (see Fig. 2). The tangent and normal components of any vector $\boldsymbol{u} \in \mathbb{R}^3$ will be denoted by $\boldsymbol{u}'$ and $u_n\boldsymbol{n}$, respectively, where $u_n = (\boldsymbol{u}, \boldsymbol{n})$ and $\boldsymbol{u}' = \boldsymbol{u} - u_n\boldsymbol{n}$, and its direction by $\hat{\boldsymbol{u}} = \frac{\boldsymbol{u}}{|\boldsymbol{u}|}$ (provided of course that $|\boldsymbol{u}| \neq 0$).

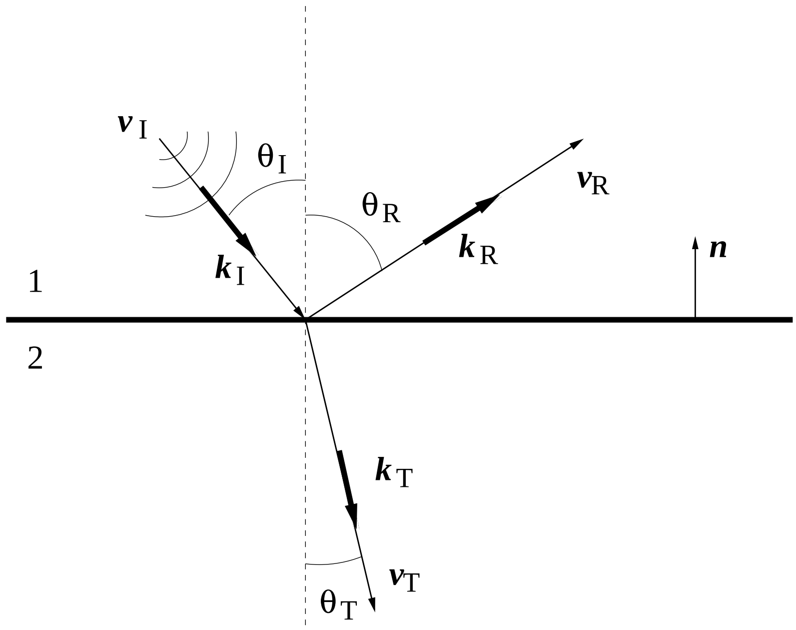

Figure 2: Air/water interface.

Q1. What are the boundary conditions to be satisfied by the velocity $\boldsymbol{v}_j$ and pressure fields p_j of both media on Σ?

Q2. Write the velocity fields $\boldsymbol{v}_j$ in both media as the combination of incident, reflected and transmitted fields.

Q3. What are the corresponding pressure fields p_j?

Q4. Derive Snell-Descartes law, and compute the reflection and transmission coefficients of the interface.

Q5. How are these coefficients modified if the incident acoustic wave impinges Σ from the water (upward on Fig. 2)?

Q6. Discuss the transmission of an acoustic signal from the air into water with the data $c_1 = 340$ m/s, $\varrho_1 = 1.3$ kg/m^3 (air), $c_2 = 1500$ m/s, $\varrho_2 = 1000$ kg/m^3 (water).

Q7. To go further (optional) *Compute the reflection/transmission coefficients for the normal acoustic intensities.*

Exercice 3: Elastic plane waves in an half-space

We consider an elastic plane wave $\boldsymbol{w} = \hat{\boldsymbol{d}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}$ with circular frequency ω, wave vector $\boldsymbol{k}$, and polarization $\hat{\boldsymbol{d}}$, such that $|\hat{\boldsymbol{d}}| = 1$, travelling in an half space $\Omega = \{\boldsymbol{x} = (x, y, z);\ \boldsymbol{x}\cdot\boldsymbol{n} = z < 0\}$ where $\boldsymbol{n}$ is the outward unit normal to the boundary $\Sigma = \{\boldsymbol{x} \in \mathbb{R}^3;\ z = 0\}$. It is constituted by an homogeneous, isotropic elastic material with density ϱ and

Lamé moduli λ and μ. The Cauchy stress tensor for the displacement field $\boldsymbol{u}$ is given by $\boldsymbol{\sigma}(\boldsymbol{u}) = \lambda \mathrm{tr}\boldsymbol{\epsilon}(\boldsymbol{u})\boldsymbol{I}_d + 2\mu\boldsymbol{\epsilon}(\boldsymbol{u})$ where $\boldsymbol{I}_d$ is the identity matrix and $\boldsymbol{\epsilon}(\boldsymbol{u}) = \boldsymbol{\nabla} \otimes_s \boldsymbol{u} = \partial_x \boldsymbol{u} \otimes_s \boldsymbol{e}_x + \partial_y \boldsymbol{u} \otimes_s \boldsymbol{e}_y + \partial_z \boldsymbol{u} \otimes_s \boldsymbol{e}_z$ ($\boldsymbol{e}_z = \boldsymbol{n}$) is the linearized stress tensor. The tangent and normal components on Σ of any vector $\boldsymbol{u} \in \mathbb{R}^3$ will be denoted by $\boldsymbol{u}'$ and $u_n\boldsymbol{n}$, respectively, where $u_n = \boldsymbol{u} \cdot \boldsymbol{n}$ and $\boldsymbol{u}' = \boldsymbol{u} - u_n\boldsymbol{n}$, and its direction by $\hat{\boldsymbol{u}} = \frac{\boldsymbol{u}}{|\boldsymbol{u}|}$. For example, $z = x_n$ with these notations.

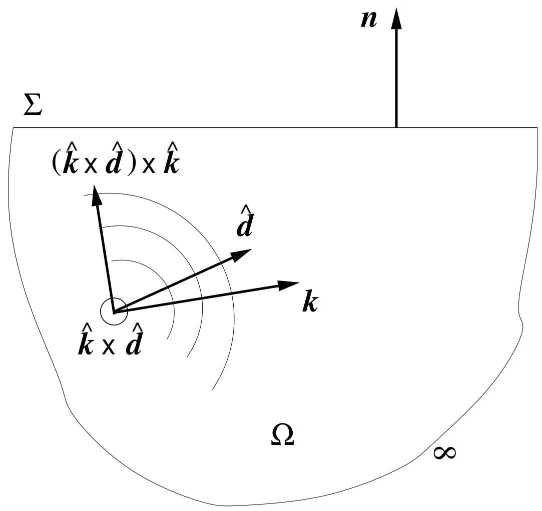

Figure 3: Elastic waves with polarization $\hat{\boldsymbol{d}}$ and wave vector $\boldsymbol{k}$ in half-space.

Q1. What are the conditions on $\hat{\boldsymbol{d}}$ and $\boldsymbol{k}$ ensuring that $\boldsymbol{w}$ is a solution of the elastic wave equation $\mathbf{Div}\boldsymbol{\sigma}(\boldsymbol{w}) = \varrho\partial_t^2\boldsymbol{w}$? [Hint: express the wave equation in the direct orthogonal triplet $(\hat{\boldsymbol{k}}, (\hat{\boldsymbol{k}} \times \hat{\boldsymbol{d}}) \times \hat{\boldsymbol{k}}, \hat{\boldsymbol{k}} \times \hat{\boldsymbol{d}})$, see Fig. 3]

Q2. Compute the stress vector $\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{w}) = \boldsymbol{\sigma}(\boldsymbol{w})\boldsymbol{n}$ exerted by $\boldsymbol{w}$ on Σ.

We now consider an incident plane wave $\boldsymbol{u}_{\mathrm{I}} = U_{\mathrm{I}}\hat{\boldsymbol{d}}_{\mathrm{I}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}_{\mathrm{I}}\cdot\boldsymbol{x}-\omega t)}$ with polarization $\hat{\boldsymbol{d}}_{\mathrm{I}}$ impinging the flat horizontal boundary Σ which is reflected as the plane waves $\boldsymbol{u}_{\mathrm{R}} = U_{\mathrm{R}}\hat{\boldsymbol{d}}_{\mathrm{R}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}_{\mathrm{R}}\cdot\boldsymbol{x}-\omega t)}$ such that a traction-free boundary condition holds on Σ.

Q3. Describe the reflected waves $\boldsymbol{u}_{\mathrm{R}}$ for the possible different incident waves.

Q4. Derive the Snell-Descartes law of refraction applicable to the elastic half-space.

Q5. Derive the reflection coefficients.

Q6. Discuss the critical incidence for total reflection with incident shear waves.

Exercice 4: Transmission of plane waves at the interface between water and solid

We finally consider the transmission of plane waves through the boundary $\Sigma = \{\boldsymbol{x} \in \mathbb{R}^3;\, z = 0\}$ between:

- a fluid medium $\Omega_1 = \{\boldsymbol{x} \in \mathbb{R}^3;\, z > 0\}$ for which $\mu_1 = 0$ and the compressibility is K_1 such that $c_1 = \sqrt{K_1/\varrho_1}$;
- and a solid medium $\Omega_2 = \{\boldsymbol{x} \in \mathbb{R}^3;\, z < 0\}$ with density ϱ_2 and Lamé moduli λ_2 and μ_2, wave celerities $c_{\mathrm{P}} = \sqrt{(\lambda_2 + 2\mu_2)/\varrho_2}$ and $c_{\mathrm{S}} = \sqrt{\mu_2/\varrho_2}$.

Let $\boldsymbol{n}$ be the upward unit normal to Σ. The tangent and normal components on Σ of any vector $\boldsymbol{u} \in \mathbb{R}^3$ will be denoted by $\boldsymbol{u}'$ and $u_n\boldsymbol{n}$, respectively, where $u_n = \boldsymbol{u}\cdot\boldsymbol{n}$ and $\boldsymbol{u}' = \boldsymbol{u} - u_n\boldsymbol{n}$, and its direction by $\hat{\boldsymbol{u}} = \frac{\boldsymbol{u}}{|\boldsymbol{u}|}$. For example, $z = x_n$ with these notations.

Q1. Describe the plane waves in the fluid medium and compute their boundary traces.

Q2. Describe the plane waves in the solid medium and compute their boundary traces.

Q3. What are the boundary conditions applicable on Σ?

Q4. Derive the Snell-Descartes law.

Q5. Derive the equations satisfied by the reflection and transmission coefficients.

Q6. Compute them for normal incidence.

Important formulas

Linearized continuity equation and balance of momentum in a fluid at rest:

$$\begin{gathered}\partial_t \varrho + \mathrm{div}_{\boldsymbol{x}}(\varrho_0 \boldsymbol{v}) = 0 \\ \varrho_0 \partial_t \boldsymbol{v} = -\mathbf{grad}_{\boldsymbol{x}} p + \boldsymbol{f}\,.\end{gathered}$$

Scalar products of trigonometric functions:

$$\int_0^1 \cos(k\pi x)\cos(k'\pi x)\mathrm{d}x = C_k \delta_{kk'}\,,\quad \int_0^1 \sin(k\pi x)\sin(k'\pi x)\mathrm{d}x = S_k \delta_{kk'}\,,$$

with $C_0 = 1$, $S_0 = 0$, and $C_k = S_k = \frac{1}{2}$ for $k \geq 1$.

Solutions

Exercice 1

1. κ is an acceleration (m/s^2), δ is a speed (m/s). κ corresponds to the inertial effect of the layer, and δ models some damping.

2. The stationary flow has $\boldsymbol{v}_{\mathrm{ref}} = \mathbf{0}$. Linearized Euler equation: $\varrho_{\mathrm{F}}\partial_t\boldsymbol{v}_{\mathrm{F}} + \mathbf{grad}_{\boldsymbol{x}} p = \boldsymbol{f}$ in Ω_{F}, constitutive equation: $\partial_t p = -\varrho_{\mathrm{F}} c_{\mathrm{F}}^2 \mathrm{div}_{\boldsymbol{x}}\boldsymbol{v}_{\mathrm{F}}$ in Ω_{F}, boundary conditions: $\partial_{\boldsymbol{n}} p = 0$ on Γ_R and $-c_{\mathrm{F}}^2\mathrm{div}_{\boldsymbol{x}}\boldsymbol{v}_{\mathrm{F}} = (\kappa+\delta\partial_t)\boldsymbol{v}_{\mathrm{F}}\cdot\boldsymbol{n}$ on Γ_Z, initial conditions: $\boldsymbol{v}_{\mathrm{F}}(\boldsymbol{x},0) = \mathbf{0}$, $p(\boldsymbol{x},0) = p_0 = 0$ (fluid at rest).

3. $\boldsymbol{f}(\boldsymbol{x},t) = \boldsymbol{F}(\boldsymbol{x},\omega)\cos\omega t$, $p(\boldsymbol{x},t) = P(\boldsymbol{x},\omega)\cos\omega t$, and then:

$$\left\{\begin{aligned} \Delta_{\boldsymbol{x}} P + \frac{\omega^2}{c_{\mathrm{F}}^2} P &= \mathrm{div}_{\boldsymbol{x}}\boldsymbol{F} && \text{in } \Omega_{\mathrm{F}}, \\ \partial_{\boldsymbol{n}} P &= 0 && \text{on } \Gamma_R, \\ \mathrm{i}\omega\mathcal{Z}(\omega)\partial_{\boldsymbol{n}} P &= \omega^2 P && \text{on } \Gamma_Z. \end{aligned}\right.$$

4. Find P such that:

$$-\omega^2\int_{\Omega_{\mathrm{F}}}\frac{PQ}{c_{\mathrm{F}}^2}\,\mathrm{d}\boldsymbol{x} + \int_{\Omega_{\mathrm{F}}}\boldsymbol{\nabla}_{\boldsymbol{x}} P\cdot\boldsymbol{\nabla}_{\boldsymbol{x}} Q\,\mathrm{d}\boldsymbol{x} + \mathrm{i}\omega\int_{\Gamma_Z}\frac{PQ}{\mathcal{Z}(\omega)}\mathrm{d}\sigma(\boldsymbol{x}) = -\int_{\Omega_{\mathrm{F}}} Q\,\mathrm{div}_{\boldsymbol{x}}\boldsymbol{F}\,\mathrm{d}\boldsymbol{x}\,,\quad \forall Q\in\mathbb{V}_0\,,$$

where:

$$\mathbb{V}_0 = \left\{Q(\boldsymbol{x}),\,\boldsymbol{x}\in\Omega_{\mathrm{F}};\,\int_{\Omega_{\mathrm{F}}}\left(\|\boldsymbol{\nabla} Q\|^2 + \frac{\omega_{\mathrm{ref}}^2}{c_{\mathrm{F}}^2}\|Q\|^2\right)\mathrm{d}\boldsymbol{x} < +\infty\right\}.$$

5. $\mathcal{Z}(\omega)\equiv\infty$ for a rigid, non-absorbing wall. Then the acoustic modes satisfy:

$$\left\{\begin{aligned} \Delta_{\boldsymbol{x}} P + \frac{\omega^2}{c_{\mathrm{F}}^2} P &= 0 && \text{in } \Omega_{\mathrm{F}}, \\ \partial_{\boldsymbol{n}} P &= 0 && \text{on } \Gamma_R\cup\Gamma_Z. \end{aligned}\right.$$

Choosing $P(\boldsymbol{x}) = f(x)g(y)h(z)$ yields $\Delta_{\boldsymbol{x}} P = (\frac{f''}{f}+\frac{g''}{g}+\frac{h''}{h})P$ and $\partial_{\boldsymbol{n}} P = (\delta_a(x)-\delta_0(x))\frac{f'}{f}P$ for $\boldsymbol{n} = (\pm1,0,0)$, with similar expressions for $\boldsymbol{n} = (0,\pm1,0)$ or $\boldsymbol{n} = (0,0,\pm1)$. Thus one has:

$$\frac{f''}{f}+\frac{g''}{g}+\frac{h''}{h}+\frac{\omega^2}{c_{\mathrm{F}}^2} = 0\quad\forall\boldsymbol{x}\in\Omega_{\mathrm{F}}\,,$$

with $f'(0) = f'(a) = 0$, $g'(0) = g'(b) = 0$, and $h'(0) = h'(c) = 0$. This shows that $\frac{f''}{f}$, $\frac{g''}{g}$, and $\frac{h''}{h}$ need be constant, and one is led to three boundary-value problems:

$$f''(x) = -\alpha^2 f(x)\ \ 0<x<a\,,\ \ f'(0) = f'(a) = 0\,,$$

$$g''(y) = -\beta^2 g(y)\ \ 0<y<b\,,\ \ g'(0) = g'(b) = 0\,,$$

and:

$$h''(z) = -\gamma^2 h(z)\ \ 0<z<c\,,\ \ h'(0) = h'(c) = 0\,,$$

such that:

$$\alpha^2+\beta^2+\gamma^2 = \frac{\omega^2}{c_{\mathrm{F}}^2}\,.$$

These problems have non-trivial solutions $f_l(x) = F(e^{\mathrm{i}\alpha_l x}+e^{-\mathrm{i}\alpha_l x})$, $g_n(x) = G(e^{\mathrm{i}\beta_m y}+e^{-\mathrm{i}\beta_m y})$, and $h_k(z) = H(e^{\mathrm{i}\gamma_n z}+e^{-\mathrm{i}\gamma_n z})$ with $\alpha_l = \frac{l\pi}{a}$, $\beta_m = \frac{m\pi}{b}$, and $\gamma_n = \frac{n\pi}{c}$, $(l,m,n)\in\mathbb{N}^3$, from the boundary conditions. The eigenmodes are:

$$P_{lmn}(\boldsymbol{x}) = A_{lmn}\cos\left(l\pi\frac{x}{a}\right)\cos\left(m\pi\frac{y}{a}\right)\cos\left(n\pi\frac{z}{c}\right)\,,$$

where the A_{lmn}'s are derived from the normalization condition, and the associated eigenfrequencies in Hz are:

$$f_{lmn} = \frac{c_{\mathrm{F}}}{2}\sqrt{\left(\frac{l}{a}\right)^2+\left(\frac{m}{b}\right)^2+\left(\frac{n}{c}\right)^2}\,.$$

6. $P_{lmn} \in \mathbb{V}_0$; using the expansion $P(\boldsymbol{x},\omega) = \sum_{l+m+n\neq 0} X_{lmn}(\omega)P_{lmn}(\boldsymbol{x})$ and $Q \equiv P_{l'm'n'}(\boldsymbol{x})$, one has to compute:

$$\frac{1}{c_{\mathrm{F}}^2}\int_{\Omega_{\mathrm{F}}} P_{lmn}P_{l'm'n'}\mathrm{d}\boldsymbol{x} = \frac{VA_{lmn}^2}{c_{\mathrm{F}}^2}C_lC_mC_n\delta_{ll'}\delta_{mm'}\delta_{nn'}$$

and:

$$\begin{aligned}&\int_{\Omega_{\mathrm{F}}} \boldsymbol{\nabla}_{\boldsymbol{x}}P_{lmn}\cdot\boldsymbol{\nabla}_{\boldsymbol{x}}P_{l'm'n'}\mathrm{d}\boldsymbol{x}\\ &= VA_{lmn}^2\pi^2\left[S_lC_mC_n\left(\frac{l}{a}\right)^2 + C_lS_mC_n\left(\frac{m}{b}\right)^2 + C_lC_mS_n\left(\frac{n}{c}\right)^2\right]\delta_{ll'}\delta_{mm'}\delta_{nn'}\\ &:= K_{lmn}\delta_{ll'}\delta_{mm'}\delta_{nn'}\,,\end{aligned}$$

where $V = abc$ is the volume of the cavity. Choosing a "mass" normalization $VA_{lmn}^2C_lC_mC_n = c_{\mathrm{F}}^2$, one has:

$$\begin{aligned}K_{lmn} &= c_{\mathrm{F}}^2\pi^2\left[\frac{S_l}{C_l}\left(\frac{l}{a}\right)^2 + \frac{S_m}{C_m}\left(\frac{m}{b}\right)^2 + \frac{S_n}{C_n}\left(\frac{n}{c}\right)^2\right]\\ &= \omega_{lmn}^2\,,\end{aligned}$$

where $\omega_{lmn} = 2\pi f_{lmn}$. At last:

$$\int_{\Gamma_Z} P_{lmn}P_{l'm'n'}\mathrm{d}\sigma(\boldsymbol{x}) = (-1)^{n+n'}A_{lmn}A_{l'mn}abC_lC_m\delta_{ll'}\delta_{mm'}\,,$$

and one is led to a *coupled* system for the dofs X_{lmn}:

$$(\omega_{lmn}^2 - \omega^2)X_{lmn} + \frac{\mathrm{i}\omega c_{\mathrm{F}}^2}{c\mathcal{Z}(\omega)}\sum_{l+m+n'\neq 0}\frac{(-1)^{n+n'}}{\sqrt{C_nC_{n'}}}X_{lmn'} = F_{lmn}$$

for $l+m+n\neq 0$, with $F_{lmn}(\omega) = -\int_{\Omega_{\mathrm{F}}} P_{lmn}\mathrm{div}_{\boldsymbol{x}}\boldsymbol{F}\,\mathrm{d}\boldsymbol{x}$.

7. The above system exhibits the mass and stiffness matrices of the cavity with rigid walls, $\mathbf{M} = \mathbf{I}$ (this is the normalization chosen above), $\mathbf{K} = \mathrm{diag}\{\omega_{lmn}^2\}$, and an added impedance effect of the coating characterized by equivalent frequency-dependent "mass" and "damping" matrices $\mathbf{M}_Z(\omega)$ and $\mathbf{D}_Z(\omega)$, respectively. The latter read in the eigenmode basis:

$$\begin{aligned}[\mathbf{M}_Z(\omega)]_{lmn,l'm'n'} &= \frac{\kappa c_{\mathrm{F}}^2}{c(\kappa^2+\delta^2\omega^2)}\frac{(-1)^{n+n'}}{\sqrt{C_nC_{n'}}}\delta_{ll'}\delta_{mm'}\,,\\ [\mathbf{D}_Z(\omega)]_{lmn,l'm'n'} &= \frac{\delta c_{\mathrm{F}}^2\omega^2}{c(\kappa^2+\delta^2\omega^2)}\frac{(-1)^{n+n'}}{\sqrt{C_nC_{n'}}}\delta_{ll'}\delta_{mm'}\,,\end{aligned}$$

such that the vector $\mathbf{X}(\omega) = (X_{lmn}(\omega))_{l+m+n>0}$ of the dofs satisfies:

$$[-\omega^2(\mathbf{I}+\mathbf{M}_Z(\omega)) + \mathrm{i}\omega\mathbf{D}_Z(\omega) + \mathbf{K}]\mathbf{X}(\omega) = \mathbf{F}(\omega)\,.$$

The coating couples the eigenmodes of the uncoated rigid cavity, which are otherwise uncoupled. If, say, $c \gg \sqrt{ab}$ (a rigid tube), the added impedance becomes negligible and the dofs decouple: this is the case of an acoustic wave guide.

8. Those *complex modes* are non-zero solutions of:

$$\left\{\begin{aligned}\Delta_{\boldsymbol{x}}\tilde{P} + \frac{\omega^2}{c_{\mathrm{F}}^2}\tilde{P} &= 0 && \text{in } \Omega_{\mathrm{F}},\\ \partial_{\boldsymbol{n}}\tilde{P} &= 0 && \text{on } \Gamma_R,\\ \mathrm{i}\omega\mathcal{Z}(\omega)\partial_{\boldsymbol{n}}\tilde{P} &= \omega^2\tilde{P} && \text{on } \Gamma_Z.\end{aligned}\right.$$

By the method of separation of variable $\tilde{P}(\boldsymbol{x}) = f(x)g(y)\tilde{h}(z)$, leading to the same differential equation as for the real eigenmodes (see question 5) with $f'(0) = f'(a) = 0$, $g'(0) = g'(b) = 0$, $\tilde{h}'(0) = 0$, but $\mathcal{Z}(\omega)\tilde{h}'(c) = -\mathrm{i}\omega\tilde{h}(c)$. This yields the three boundary-value problems:

$$f''(x) = -\alpha^2 f(x)\ \ 0<x<a\,,\ \ f'(0) = f'(a) = 0\,,$$

$$g''(y) = -\beta^2 g(y) \;\; 0 < y < b\,, \;\; g'(0) = g'(b) = 0\,,$$

and now:

$$\tilde{h}''(z) = -\gamma^2 \tilde{h}(z) \;\; 0 < z < c\,, \;\; \tilde{h}'(0) = 0\,, \;\; \mathcal{Z}(\omega)\tilde{h}'(c) = -\mathrm{i}\omega\tilde{h}(c)\,,$$

such that $\gamma^2 = \frac{\omega^2}{c_{\mathrm{F}}^2} - \alpha^2 - \beta^2$; here γ may be either real or complex. The first two problems have the same non-trivial solutions as above (see question 5), and the third one has solutions $\tilde{h}_n(z) = H(e^{\mathrm{i}\gamma_n\pi\frac{z}{c}} + e^{-\mathrm{i}\gamma_n\pi\frac{z}{c}})$ with:

$$\gamma_n\pi\tan\gamma_n\pi = \frac{\mathrm{i}\omega c}{\mathcal{Z}(\omega)} = -\frac{\omega^2 c}{\kappa + \mathrm{i}\omega\delta}\,.$$

The eigenmodes are now $\tilde{P}_{lmn}(\boldsymbol{x}) = \tilde{A}_{lmn}\cos(l\pi\frac{x}{a})\cos(m\pi\frac{y}{a})\cos(\gamma_n\pi\frac{z}{c})$ where the $\tilde{A}_{lmn}$'s are derived from the normalization condition, and the associated (complex) eigenfrequencies are:

$$\tilde{f}_{lmn} = \frac{c_{\mathrm{F}}}{2}\sqrt{\left(\frac{l}{a}\right)^2 + \left(\frac{m}{b}\right)^2 + \left(\frac{\gamma_n}{c}\right)^2}\,.$$

The Galerkin formulation leads to the following *uncoupled* equations for the dof's $\tilde{X}_{lmn}$:

$$(\tilde{\omega}_{lmn}^2 - \omega^2)\tilde{X}_{lmn} = \tilde{F}_{lmn}\,,$$

where $\tilde{\omega}_{lmn} = 2\pi\tilde{f}_{lmn} \in \mathbb{C}$.

Exercice 2

1. $\boldsymbol{v}_1\cdot\boldsymbol{n} = \boldsymbol{v}_2\cdot\boldsymbol{n}$, $p_1 = p_2$ on Σ.
2. One has:

$$\boldsymbol{v}_1 = \boldsymbol{v}_{\mathrm{I}} + \boldsymbol{v}_{\mathrm{R}} = c_1 e^{\mathrm{i}\omega t}(\hat{\boldsymbol{k}}_{\mathrm{I}}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_{\mathrm{I}}\cdot\boldsymbol{x}} + R\hat{\boldsymbol{k}}_{\mathrm{R}}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_{\mathrm{R}}\cdot\boldsymbol{x}})\,,$$
$$\boldsymbol{v}_2 = \boldsymbol{v}_{\mathrm{T}} = T\times c_2\hat{\boldsymbol{k}}_{\mathrm{T}}e^{\mathrm{i}(\omega t - \boldsymbol{k}_{\mathrm{T}}\cdot\boldsymbol{x})}\,,$$

   where R/T are the reflection/transmission coefficients of the interface.
3. Linearized continuity equation: $\frac{\mathrm{d}}{\mathrm{d}t}(\frac{p_j}{c_j^2}) = -\varrho_j\mathrm{div}\boldsymbol{v}_j$, hence:

$$p_1 = \varrho_1 c_1^2 e^{\mathrm{i}\omega t}(e^{-\mathrm{i}\boldsymbol{k}_{\mathrm{I}}\cdot\boldsymbol{x}} + Re^{-\mathrm{i}\boldsymbol{k}_{\mathrm{R}}\cdot\boldsymbol{x}})\,, \;\; p_2 = \varrho_2 c_2^2 T e^{\mathrm{i}(\omega t - \boldsymbol{k}_{\mathrm{T}}\cdot\boldsymbol{x})}\,,$$

   owing to $|\boldsymbol{k}_{\mathrm{I}}| = |\boldsymbol{k}_{\mathrm{R}}| = \frac{\omega}{c_1}$, $|\boldsymbol{k}_{\mathrm{T}}| = \frac{\omega}{c_2}$, and $\mathrm{div}(\boldsymbol{d}\,e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}) = \mathrm{i}(\boldsymbol{d}\cdot\boldsymbol{k})e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}$.
4. The boundary conditions on $\Sigma = \{\boldsymbol{x}\in\mathbb{R}^3;\, x_n = 0\}$ yield:

$$c_1^2(k_{\mathrm{I}n}e^{-\mathrm{i}\boldsymbol{k}'_{\mathrm{I}}\cdot\boldsymbol{x}'} + Rk_{\mathrm{R}n}e^{-\mathrm{i}\boldsymbol{k}'_{\mathrm{R}}\cdot\boldsymbol{x}'}) = c_2^2 T k_{\mathrm{T}n}e^{-\mathrm{i}\boldsymbol{k}'_{\mathrm{T}}\cdot\boldsymbol{x}'}\,, \tag{1}$$
$$\varrho_1 c_1^2(e^{-\mathrm{i}\boldsymbol{k}'_{\mathrm{I}}\cdot\boldsymbol{x}'} + Re^{-\mathrm{i}\boldsymbol{k}'_{\mathrm{R}}\cdot\boldsymbol{x}'}) = \varrho_2 c_2^2 T e^{-\mathrm{i}\boldsymbol{k}'_{\mathrm{T}}\cdot\boldsymbol{x}'}\,, \tag{2}$$

   $\forall\boldsymbol{x}'\in\Sigma$. Thus $\boldsymbol{k}'_{\mathrm{I}} = \boldsymbol{k}'_{\mathrm{R}} = \boldsymbol{k}'_{\mathrm{T}} \equiv \boldsymbol{k}'$ (Snell-Descartes law), $-k_{\mathrm{I}n} = k_{\mathrm{R}n} = k_{1n}(\boldsymbol{k}')$, $-k_{\mathrm{T}n} = k_{2n}(\boldsymbol{k}')$ where $k_{jn}(\boldsymbol{k}') = (\frac{\omega^2}{c_j^2} - |\boldsymbol{k}'|^2)^{\frac{1}{2}}$, and:

$$R(\boldsymbol{k}') = \frac{\varrho_2 k_{1n}(\boldsymbol{k}') - \varrho_1 k_{2n}(\boldsymbol{k}')}{\varrho_2 k_{1n}(\boldsymbol{k}') + \varrho_1 k_{2n}(\boldsymbol{k}')}\,, \;\; T(\boldsymbol{k}') = 2\left(\frac{\varrho_1 c_1^2}{\varrho_2 c_2^2}\right)\frac{\varrho_2 k_{1n}(\boldsymbol{k}')}{\varrho_2 k_{1n}(\boldsymbol{k}') + \varrho_1 k_{2n}(\boldsymbol{k}')}\,.$$

5. Obviously $R_{\mathrm{water}}(\boldsymbol{k}') = -R_{\mathrm{air}}(\boldsymbol{k}')$ and $T_{\mathrm{water}}(\boldsymbol{k}') = \frac{\varrho_2 c_2^4 k_{2n}(\boldsymbol{k}')}{\varrho_1 c_1^4 k_{1n}(\boldsymbol{k}')}T_{\mathrm{air}}(\boldsymbol{k}')$ if subscripts 1 and 2 are switched.
6. For a normal incidence $k_{jn} = \frac{\omega}{c_j}$, $j = 1, 2$, and:

$$R(\boldsymbol{0}) = \frac{\varrho_2 c_2 - \varrho_1 c_1}{\varrho_1 c_1 + \varrho_2 c_2} \simeq 1\,, \;\; T(\boldsymbol{0}) = 2\left(\frac{c_1}{c_2}\right)\frac{\varrho_1 c_1}{\varrho_1 c_1 + \varrho_2 c_2} \simeq 10^{-4}!!$$

   Also above the critical incidence angle $\theta_{\mathrm{I}} > \theta_c = \arcsin\frac{c_1}{c_2} \simeq 13°$ no sound at all is transmitted (total reflection).
7. $\mathcal{R}(\boldsymbol{k}') = -\frac{p_{\mathrm{R}}\boldsymbol{v}_{\mathrm{R}}\cdot\boldsymbol{n}}{p_{\mathrm{I}}\boldsymbol{v}_{\mathrm{I}}\cdot\boldsymbol{n}}$ and $\mathcal{T}(\boldsymbol{k}') = \frac{p_{\mathrm{T}}\boldsymbol{v}_{\mathrm{T}}\cdot\boldsymbol{n}}{p_{\mathrm{I}}\boldsymbol{v}_{\mathrm{I}}\cdot\boldsymbol{n}}$, that is:

$$\mathcal{T}(\boldsymbol{k}') = \frac{\varrho_2 c_2^4 k_{2n}(\boldsymbol{k}')}{\varrho_1 c_1^4 k_{1n}(\boldsymbol{k}')}T(\boldsymbol{k}')^2 = \frac{4\varrho_1 k_{2n}(\boldsymbol{k}')\varrho_2 k_{1n}(\boldsymbol{k}')}{(\varrho_2 k_{1n}(\boldsymbol{k}') + \varrho_1 k_{2n}(\boldsymbol{k}'))^2}$$

   and $\mathcal{R}(\boldsymbol{k}') = R(\boldsymbol{k}')^2$; one can check that $\mathcal{R}(\boldsymbol{k}') + \mathcal{T}(\boldsymbol{k}') = 1$ (conservation of energy flux).

Exercice 3

1. For a plane wave the linearized strain tensor is $\boldsymbol{\epsilon}(\boldsymbol{w}) = \mathrm{i}\boldsymbol{k} \otimes_s \hat{\boldsymbol{d}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$, so that $\boldsymbol{\sigma}(\boldsymbol{w}) = \mathrm{i}[\lambda(\hat{\boldsymbol{d}}\cdot\boldsymbol{k})\boldsymbol{I}_d + 2\mu\boldsymbol{k}\otimes_s\hat{\boldsymbol{d}}]\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$. Thus $\varrho\partial_t^2\boldsymbol{w} - \mathbf{Div}\boldsymbol{\sigma}(\boldsymbol{w}) = [\varrho\omega^2\hat{\boldsymbol{d}} - (\lambda+\mu)|\boldsymbol{k}|^2(\hat{\boldsymbol{d}}\cdot\hat{\boldsymbol{k}})\hat{\boldsymbol{k}} - \mu|\boldsymbol{k}|^2\hat{\boldsymbol{d}}]\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$. But $\hat{\boldsymbol{d}} = (\hat{\boldsymbol{k}}\times\hat{\boldsymbol{d}})\times\hat{\boldsymbol{k}} + (\hat{\boldsymbol{d}}\cdot\hat{\boldsymbol{k}})\hat{\boldsymbol{k}}$ if one forms a direct trihedron attached to the plane $(\hat{\boldsymbol{d}},\hat{\boldsymbol{k}})$, and therefore the elastic wave equation ends up with:
$$(\omega^2 - c_{\mathrm{S}}^2|\boldsymbol{k}|^2)(\hat{\boldsymbol{k}}\times\hat{\boldsymbol{d}})\times\hat{\boldsymbol{k}} + (\omega^2 - c_{\mathrm{P}}^2|\boldsymbol{k}|^2)(\hat{\boldsymbol{d}}\cdot\hat{\boldsymbol{k}})\hat{\boldsymbol{k}} = \mathbf{0}\,,$$
where $c_{\mathrm{S}}^2 = \frac{\mu}{\varrho}$ and $c_{\mathrm{P}}^2 = \frac{\lambda+2\mu}{\varrho}$, such that $c_{\mathrm{P}} > c_{\mathrm{S}}$. Two cases arise:
   (a) either $|\boldsymbol{k}|^2 = \frac{\omega^2}{c_{\mathrm{S}}^2} = k_{\mathrm{S}}^2$ and $\hat{\boldsymbol{d}}\cdot\hat{\boldsymbol{k}} = 0$, that is $\hat{\boldsymbol{d}}\perp\hat{\boldsymbol{k}}$;
   (b) or $|\boldsymbol{k}|^2 = \frac{\omega^2}{c_{\mathrm{P}}^2} = k_{\mathrm{P}}^2$ and $\hat{\boldsymbol{k}}\times\hat{\boldsymbol{d}} = \mathbf{0}$, that is $\hat{\boldsymbol{d}}\parallel\hat{\boldsymbol{k}}$.

   The first case corresponds to transverse (shear) waves at the celerity c_{S}, and the second case corresponds to longitudinal (traction) waves at the celerity c_{P}. Since $c_{\mathrm{P}} > c_{\mathrm{S}}$, the latter are also called primary waves (hence the P subscript) and the former secondary waves.

2. $\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{w}) = \mathrm{i}[\lambda(\hat{\boldsymbol{d}}\cdot\boldsymbol{k})\boldsymbol{n} + \mu(\boldsymbol{k}\cdot\boldsymbol{n})\hat{\boldsymbol{d}} + \mu(\hat{\boldsymbol{d}}\cdot\boldsymbol{n})\boldsymbol{k}]\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}'\cdot\boldsymbol{x}'-\omega t)}$, because $x_n = \boldsymbol{x}\cdot\boldsymbol{n} = 0$ on Σ. Let $\hat{\boldsymbol{k}} = \boldsymbol{\xi}' + \xi_n\boldsymbol{n}$ such that $\xi_n^2 = 1 - |\boldsymbol{\xi}'|^2$. If $|\boldsymbol{k}| = k_{\mathrm{P}}$ then $\hat{\boldsymbol{d}} = \hat{\boldsymbol{k}}$ and:
$$\begin{aligned}\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{w}_{\mathrm{P}}) &= \mathrm{i}\mu k_{\mathrm{P}}[(\kappa^2-2)\boldsymbol{n} + 2\xi_n\hat{\boldsymbol{k}}]\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}'\cdot\boldsymbol{x}'-\omega t)} \\ &= \mathrm{i}\mu k_{\mathrm{P}}\left[(\kappa^2 - 2|\boldsymbol{\xi}'|^2)\boldsymbol{n} + 2\xi_n|\boldsymbol{\xi}'|\boldsymbol{\tau}\right]\mathrm{e}^{\mathrm{i}(k_{\mathrm{P}}\boldsymbol{\xi}'\cdot\boldsymbol{x}'-\omega t)}\,,\end{aligned}$$
with $\kappa = \frac{c_{\mathrm{P}}}{c_{\mathrm{S}}}$ and $\boldsymbol{\tau} = \hat{\boldsymbol{\xi}}' = \frac{\boldsymbol{\xi}'}{|\boldsymbol{\xi}'|}$. But if $|\boldsymbol{k}| = k_{\mathrm{S}}$ then:
$$\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{w}_{\mathrm{S}}) = \mathrm{i}\mu k_{\mathrm{S}}[\xi_n\hat{\boldsymbol{d}} + \hat{d}_n\hat{\boldsymbol{k}}]\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}'\cdot\boldsymbol{x}'-\omega t)}$$
because $\hat{\boldsymbol{d}}\perp\hat{\boldsymbol{k}}$. Two sub-cases arise:
   (a) either $\hat{\boldsymbol{d}} = \frac{\hat{\boldsymbol{k}}\times\boldsymbol{n}}{|\hat{\boldsymbol{k}}\times\boldsymbol{n}|} = \boldsymbol{\tau}\times\boldsymbol{n}$, in which case $\hat{d}_n = 0$ and the shear waves are so-called SH waves polarized perpendicularly to the plane of propagation–hence horizontally:
$$\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{w}_{\mathrm{SH}}) = \mathrm{i}\mu k_{\mathrm{S}}\xi_n(\boldsymbol{\tau}\times\boldsymbol{n})\,\mathrm{e}^{\mathrm{i}(k_{\mathrm{S}}\boldsymbol{\xi}'\cdot\boldsymbol{x}'-\omega t)}\,;$$
   (b) or $\hat{\boldsymbol{d}} = \frac{(\hat{\boldsymbol{k}}\times\boldsymbol{n})\times\hat{\boldsymbol{k}}}{|(\hat{\boldsymbol{k}}\times\boldsymbol{n})\times\hat{\boldsymbol{k}}|} = |\boldsymbol{\xi}'|\boldsymbol{n} - \xi_n\boldsymbol{\tau}$, in which case $\hat{d}_n = |\boldsymbol{\xi}'|$ and the shear waves are so-called SV waves polarized perpendicularly to $\boldsymbol{k}$ in the plane of propagation $(\boldsymbol{\tau},\boldsymbol{n})$:
$$\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{w}_{\mathrm{SV}}) = \mathrm{i}\mu k_{\mathrm{S}}\left[2\xi_n|\boldsymbol{\xi}'|\boldsymbol{n} + (|\boldsymbol{\xi}'|^2 - \xi_n^2)\boldsymbol{\tau}\right]\mathrm{e}^{\mathrm{i}(k_{\mathrm{S}}\boldsymbol{\xi}'\cdot\boldsymbol{x}'-\omega t)}\,.$$

3. Traction-free boundary condition $\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{u}_{\mathrm{I}} + \boldsymbol{u}_{\mathrm{R}}) = \mathbf{0}$, which applies for all $\boldsymbol{x}'\in\Sigma$. Besides:
$$\begin{aligned}\boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{u}_{\mathrm{R}}) = \;& \mathrm{i}\mu k_{\mathrm{P}}R_{\mathrm{P}}\left[(\kappa^2 - 2|\boldsymbol{\xi}'_{\mathrm{RP}}|^2)\boldsymbol{n} + 2\xi_{\mathrm{RP}n}|\boldsymbol{\xi}'_{\mathrm{RP}}|\boldsymbol{\tau}\right]\mathrm{e}^{\mathrm{i}(k_{\mathrm{P}}\boldsymbol{\xi}'_{\mathrm{RP}}\cdot\boldsymbol{x}'-\omega t)} \\ &+ \mathrm{i}\mu k_{\mathrm{S}}R_{\mathrm{SV}}\left[2\xi_{\mathrm{RS}n}|\boldsymbol{\xi}'_{\mathrm{RS}}|\boldsymbol{n} + (|\boldsymbol{\xi}'_{\mathrm{RS}}|^2 - \xi_{\mathrm{RS}n}^2)\boldsymbol{\tau}\right]\mathrm{e}^{\mathrm{i}(k_{\mathrm{S}}\boldsymbol{\xi}'_{\mathrm{RS}}\cdot\boldsymbol{x}'-\omega t)} \\ &+ \mathrm{i}\mu k_{\mathrm{S}}R_{\mathrm{SH}}\xi_{\mathrm{RS}n}(\boldsymbol{\tau}\times\boldsymbol{n})\,\mathrm{e}^{\mathrm{i}(k_{\mathrm{S}}\boldsymbol{\xi}'_{\mathrm{RS}}\cdot\boldsymbol{x}'-\omega t)}\end{aligned}$$
We see that if the incident wave $\boldsymbol{u}_{\mathrm{I}}$ is polarized along $\boldsymbol{\tau}\times\boldsymbol{n}$ then it gives rise to reflected SH waves solely, whereas incident waves polarized in the plane $(\boldsymbol{\tau},\boldsymbol{n})$ give rise to both P and SV waves.

4. Therefore, all phase terms $\mathrm{i}k_\alpha\boldsymbol{\xi}'\cdot\boldsymbol{x}'$ are equal for either $\alpha = \mathrm{P}$ or $\alpha = \mathrm{S}$, that is:
$$k_\alpha\boldsymbol{\xi}'_{\mathrm{I}\alpha} = k_\beta\boldsymbol{\xi}'_{\mathrm{R}\beta} := \boldsymbol{k}'\,,\quad \alpha,\beta = \mathrm{P},\mathrm{S}\,,$$
and consequently (the reflected waves are going downward if the incident waves are going upward such that $\xi_{\mathrm{I}\alpha n} > 0$):
$$\xi_{\mathrm{I}\alpha n}(\boldsymbol{k}') = \sqrt{1 - \frac{|\boldsymbol{k}'|^2}{k_\alpha^2}}\,,\quad \xi_{\mathrm{R}\beta n}(\boldsymbol{k}') = -\sqrt{1 - \frac{|\boldsymbol{k}'|^2}{k_\beta^2}}\,.$$

5. If the incident wave is polarized along $\boldsymbol{\tau} \times \boldsymbol{n}$ one has $\mathrm{i}\mu k_{\mathrm{S}} \left(\xi_{\mathrm{IS}n} + R_{\mathrm{SH}}(\boldsymbol{k}')\xi_{\mathrm{RS}n}(\boldsymbol{k}') \right) = 0$, or:

$$R_{\mathrm{SH}}(\boldsymbol{k}') = 1\,.$$

That is, SH waves are totally reflected independently of the other polarization modes. If the incident wave is a P wave such that $|\boldsymbol{k}| = k_{\mathrm{P}}$, one has:

$$\begin{aligned}\mathrm{i}\mu k_{\mathrm{P}} \left[(\kappa^2 - 2|\boldsymbol{\xi}'_{\mathrm{IP}}|^2)\boldsymbol{n} + 2\xi_{\mathrm{IP}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\mathrm{IP}}|\boldsymbol{\tau} \right] = &- \mathrm{i}\mu k_{\mathrm{P}} R_{\mathrm{P}}(\boldsymbol{k}') \left[(\kappa^2 - 2|\boldsymbol{\xi}'_{\mathrm{RP}}|^2)\boldsymbol{n} + 2\xi_{\mathrm{RP}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\mathrm{RP}}|\boldsymbol{\tau} \right] \\ &- \mathrm{i}\mu k_{\mathrm{S}} R_{\mathrm{SV}}(\boldsymbol{k}') \left[2\xi_{\mathrm{RS}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\mathrm{RS}}|\boldsymbol{n} + (|\boldsymbol{\xi}'_{\mathrm{RS}}|^2 - \xi^2_{\mathrm{RS}n}(\boldsymbol{k}'))\boldsymbol{\tau} \right]\,,\end{aligned}$$

or:

$$\begin{bmatrix} 2|\boldsymbol{k}'|^2 - k_{\mathrm{S}}^2 & -2k_{\mathrm{P}}\xi_{\mathrm{RS}n}(\boldsymbol{k}')|\boldsymbol{k}'| \\ 2k_{\mathrm{S}}\xi_{\mathrm{RP}n}(\boldsymbol{k}')|\boldsymbol{k}'| & 2|\boldsymbol{k}'|^2 - k_{\mathrm{S}}^2 \end{bmatrix} \begin{pmatrix} R_{\mathrm{P}}(\boldsymbol{k}') \\ R_{\mathrm{SV}}(\boldsymbol{k}') \end{pmatrix} = \begin{pmatrix} k_{\mathrm{S}}^2 - 2|\boldsymbol{k}'|^2 \\ -2k_{\mathrm{S}}\xi_{\mathrm{IP}n}(\boldsymbol{k}')|\boldsymbol{k}'| \end{pmatrix}\,.$$

If at last the incident wave is a SV wave such that $|\boldsymbol{k}| = k_{\mathrm{S}}$, one has:

$$\begin{aligned}\mathrm{i}\mu k_{\mathrm{S}} \left[2\xi_{\mathrm{IS}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\mathrm{IS}}|\boldsymbol{n} + (|\boldsymbol{\xi}'_{\mathrm{IS}}|^2 - \xi^2_{\mathrm{IS}n}(\boldsymbol{k}'))\boldsymbol{\tau} \right] = &- \mathrm{i}\mu k_{\mathrm{P}} R_{\mathrm{P}}(\boldsymbol{k}') \left[(\kappa^2 - 2|\boldsymbol{\xi}'_{\mathrm{RP}}|^2)\boldsymbol{n} + 2\xi_{\mathrm{RP}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\mathrm{RP}}|\boldsymbol{\tau} \right] \\ &- \mathrm{i}\mu k_{\mathrm{S}} R_{\mathrm{SV}}(\boldsymbol{k}') \left[2\xi_{\mathrm{RS}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\mathrm{RS}}|\boldsymbol{n} + (|\boldsymbol{\xi}'_{\mathrm{RS}}|^2 - \xi^2_{\mathrm{RS}n}(\boldsymbol{k}'))\boldsymbol{\tau} \right]\,,\end{aligned}$$

or:

$$\begin{bmatrix} 2|\boldsymbol{k}'|^2 - k_{\mathrm{S}}^2 & -2k_{\mathrm{P}}\xi_{\mathrm{RS}n}(\boldsymbol{k}')|\boldsymbol{k}'| \\ 2k_{\mathrm{S}}\xi_{\mathrm{RP}n}(\boldsymbol{k}')|\boldsymbol{k}'| & 2|\boldsymbol{k}'|^2 - k_{\mathrm{S}}^2 \end{bmatrix} \begin{pmatrix} R_{\mathrm{P}}(\boldsymbol{k}') \\ R_{\mathrm{SV}}(\boldsymbol{k}') \end{pmatrix} = \begin{pmatrix} 2k_{\mathrm{P}}\xi_{\mathrm{IS}n}(\boldsymbol{k}')|\boldsymbol{k}'| \\ k_{\mathrm{S}}^2 - 2|\boldsymbol{k}'|^2 \end{pmatrix}\,.$$

These coefficients are plotted on Fig. 4 for incident P or SV waves and a Poisson coefficient $\nu = 0.3$.

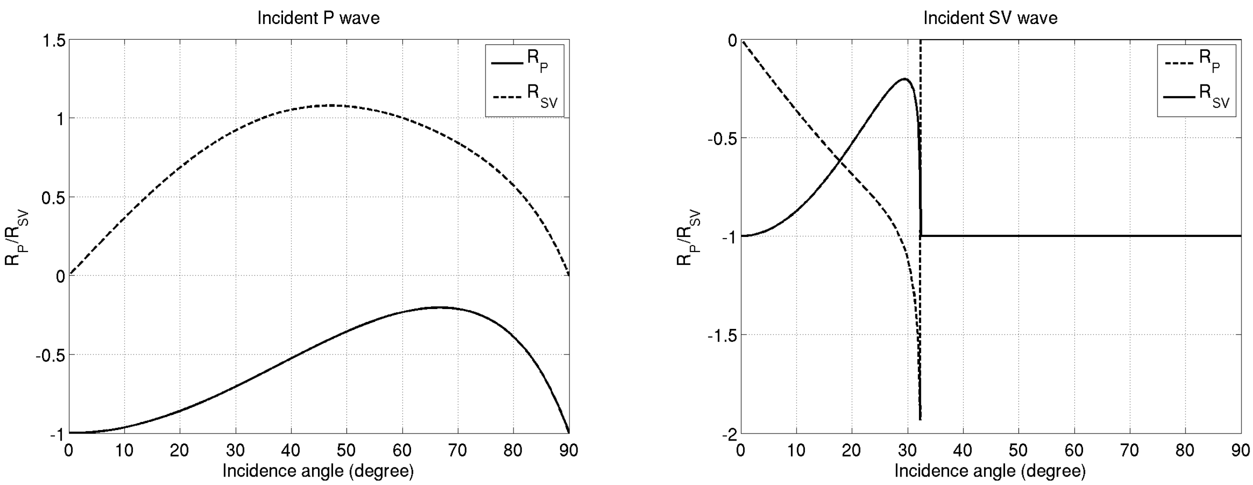

Figure 4: Reflection coefficients for incident P (left) and SV (right) waves as functions of the incidence angle $\theta = \arcsin|\boldsymbol{\xi}'|$ for $\nu = 0.3$.

6. Critical incidence with SV waves arises for $|\boldsymbol{k}'| > k_{\mathrm{P}}$, in which case $\xi_{\mathrm{RP}n}$ is purely imaginary. The reflected P waves are evanescent so that actually total SV reflection occurs: $R_{\mathrm{SV}}(|\boldsymbol{k}'| > k_{\mathrm{P}}) = -1$ while $R_{\mathrm{P}}(|\boldsymbol{k}'| > k_{\mathrm{P}}) = 0$.

Exercice 4

1. In the fluid medium $\Omega_1 = \{\boldsymbol{x};\, x_n > 0\}$, one has the boundary traces (removing the $\mathrm{e}^{-\mathrm{i}\omega t}$ terms from now on):

$$\begin{aligned}\boldsymbol{u}_{\mathrm{I}} &= \hat{\boldsymbol{k}}_{\mathrm{I}}\,\mathrm{e}^{\mathrm{i}k_1\boldsymbol{\xi}'_{\mathrm{I}}\cdot\boldsymbol{x}'}\,, & \boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{u}_{\mathrm{I}}) &= \mathrm{i}K_1k_1\boldsymbol{n}\,\mathrm{e}^{\mathrm{i}k_1\boldsymbol{\xi}'_{\mathrm{I}}\cdot\boldsymbol{x}'}\,, \\ \boldsymbol{u}_{\mathrm{R}} &= R\hat{\boldsymbol{k}}_{\mathrm{R}}\,\mathrm{e}^{\mathrm{i}k_1\boldsymbol{\xi}'_{\mathrm{R}}\cdot\boldsymbol{x}'}\,, & \boldsymbol{t}_{\boldsymbol{n}}(\boldsymbol{u}_{\mathrm{R}}) &= \mathrm{i}RK_1k_1\boldsymbol{n}\,\mathrm{e}^{\mathrm{i}k_1\boldsymbol{\xi}'_{\mathrm{R}}\cdot\boldsymbol{x}'}\,,\end{aligned}$$

where R is the reflection coefficient and $k_1 = \frac{\omega}{c_1}$.

2. In the solid medium $\Omega_2 = \{\boldsymbol{x};\, x_n < 0\}$ one has the boundary traces:

$$\boldsymbol{u}_{\mathrm{P}} = T_{\mathrm{P}}\hat{\boldsymbol{k}}_{\mathrm{P}}\,\mathrm{e}^{\mathrm{i}k_{\mathrm{P}}\boldsymbol{\xi}'_{\mathrm{P}}\cdot\boldsymbol{x}'}\,, \quad \boldsymbol{u}_{\mathrm{SV}} = T_{\mathrm{SV}}\hat{\boldsymbol{k}}_{\mathrm{SV}}\,\mathrm{e}^{\mathrm{i}k_{\mathrm{S}}\boldsymbol{\xi}'_{\mathrm{SV}}\cdot\boldsymbol{x}'}\,,$$

and:

$$\boldsymbol{t_n}(\boldsymbol{u}_{\rm P}) = {\rm i}T_{\rm P}\mu_2 k_{\rm P}\left[(\kappa^2-2|\boldsymbol{\xi}'_{\rm P}|^2)\boldsymbol{n}+2\xi_{{\rm P}n}|\boldsymbol{\xi}'_{\rm P}|\boldsymbol{\tau}\right]{\rm e}^{{\rm i}k_{\rm P}\boldsymbol{\xi}'_{\rm P}\cdot\boldsymbol{x}'}\,,$$
$$\boldsymbol{t_n}(\boldsymbol{u}_{\rm SV}) = {\rm i}T_{\rm SV}\mu_2 k_{\rm S}\left[2\xi_{{\rm S}n}|\boldsymbol{\xi}'_{\rm S}|\boldsymbol{n}+(|\boldsymbol{\xi}'_{\rm S}|^2-\xi_{{\rm S}n}^2)\boldsymbol{\tau}\right]{\rm e}^{{\rm i}k_{\rm S}\boldsymbol{\xi}'_{\rm S}\cdot\boldsymbol{x}'}\,,$$

where $T_{\rm P}$ and $T_{\rm SV}$ are the transmission coefficients of the compressional and shear waves, respectively.

3. Continuity of the normal displacements $(\boldsymbol{u}_{\rm I}+\boldsymbol{u}_{\rm R})\cdot\boldsymbol{n}=\boldsymbol{u}_{\rm T}\cdot\boldsymbol{n}$; continuity of the normal component of the stress vector $\boldsymbol{t_n}(\boldsymbol{u}_{\rm I}+\boldsymbol{u}_{\rm R})\cdot\boldsymbol{n}=\boldsymbol{t_n}(\boldsymbol{u}_{\rm T})\cdot\boldsymbol{n}$; no shear stress at the interface $\boldsymbol{t}'_{\boldsymbol{n}}(\boldsymbol{u}_{\rm T})=\boldsymbol{0}$.

4. By the Snell-Descartes law $\boldsymbol{\xi}'_{\rm I}=\boldsymbol{\xi}'_{\rm R}=\boldsymbol{\xi}'$ for the acoustic waves and therefore:

$$\xi_{{\rm R}n}(\boldsymbol{k}') = -\xi_{{\rm I}n}(\boldsymbol{k}') = \sqrt{1-\frac{|\boldsymbol{k}'|^2}{k_1^2}}\,.$$

   By the Snell-Descartes law $k_{\rm P}\boldsymbol{\xi}'_{\rm P}=k_{\rm S}\boldsymbol{\xi}'_{\rm S}=k_1\boldsymbol{\xi}'=\boldsymbol{k}'$ for the elastic waves and therefore:

$$\xi_{{\rm P}n}(\boldsymbol{k}') = -\sqrt{1-\frac{|\boldsymbol{k}'|^2}{k_{\rm P}^2}}\,,\quad \xi_{{\rm S}n}(\boldsymbol{k}') = -\sqrt{1-\frac{|\boldsymbol{k}'|^2}{k_{\rm S}^2}}\,.$$

5. The reflection and transmission coefficients as functions of $\boldsymbol{k}'$ are finally determined from the foregoing boundary conditions where $\boldsymbol{u}_{\rm T}=\boldsymbol{u}_{\rm P}+\boldsymbol{u}_{\rm SV}$:

$$\begin{gathered}\xi_{{\rm I}n}(\boldsymbol{k}')+\xi_{{\rm R}n}(\boldsymbol{k}')R(\boldsymbol{k}') = \xi_{{\rm P}n}(\boldsymbol{k}')T_{\rm P}(\boldsymbol{k}')+\xi_{{\rm S}n}(\boldsymbol{k}')T_{\rm SV}(\boldsymbol{k}')\,,\\ K_1k_1(1+R(\boldsymbol{k}')) = \mu_2k_{\rm P}(\kappa^2-2|\boldsymbol{\xi}'_{\rm P}|^2)T_{\rm P}(\boldsymbol{k}')+2\mu_2k_{\rm S}\xi_{{\rm S}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\rm S}|T_{\rm SV}(\boldsymbol{k}')\,,\\ 2\mu_2k_{\rm P}\xi_{{\rm P}n}(\boldsymbol{k}')|\boldsymbol{\xi}'_{\rm P}|T_{\rm P}(\boldsymbol{k}')+\mu_2k_{\rm S}(|\boldsymbol{\xi}'_{\rm S}|^2-\xi_{{\rm S}n}^2(\boldsymbol{k}'))T_{\rm SV}(\boldsymbol{k}')=0\,,\end{gathered}$$

   or:

$$\begin{bmatrix}-\xi_{{\rm R}n}(\boldsymbol{k}') & \xi_{{\rm P}n}(\boldsymbol{k}') & \xi_{{\rm S}n}(\boldsymbol{k}')\\ \frac{\mathcal{Z}_1}{\mathcal{Z}_{\rm S}}k_{\rm S}k_{\rm P} & 2|\boldsymbol{k}'|^2-k_{\rm S}^2 & -2k_{\rm P}\xi_{{\rm S}n}(\boldsymbol{k}')|\boldsymbol{k}'|\\ 0 & 2k_{\rm S}\xi_{{\rm P}n}(\boldsymbol{k}')|\boldsymbol{k}'| & 2|\boldsymbol{k}'|^2-k_{\rm S}^2\end{bmatrix}\begin{pmatrix}R(\boldsymbol{k}')\\ T_{\rm P}(\boldsymbol{k}')\\ T_{\rm SV}(\boldsymbol{k}')\end{pmatrix}=\begin{pmatrix}\xi_{{\rm I}n}(\boldsymbol{k}')\\ -\frac{\mathcal{Z}_1}{\mathcal{Z}_{\rm S}}k_{\rm S}k_{\rm P}\\ 0\end{pmatrix}.$$

   where $\mathcal{Z}_1=\varrho_1c_1$ is the impedance of the acoustic medium, and $\mathcal{Z}_{\rm S}=\varrho_2c_{\rm S}$ is the shear impedance of the elastic medium. These coefficients are plotted on Fig. 5 for an incident wave in the fluid medium with $\frac{\varrho_2}{\varrho_1}=3$, $\frac{c_{\rm P}}{c_1}=3$, and $\nu=0.3$ (we remind that we have $\frac{c_{\rm P}^2}{c_{\rm S}^2}=\frac{2(1-\nu)}{1-2\nu}$). Two critical incidences arise for $|\boldsymbol{k}'|>k_{\rm P}$ at first and then $|\boldsymbol{k}'|>k_{\rm S}>k_{\rm P}$, in which case $\xi_{{\rm P}n}$ and then $\xi_{{\rm S}n}$ get purely imaginary. The first critical angle is $\theta_c=\arcsin\frac{c_1}{c_{\rm P}}=19.5^\circ$ and the second critical angle is $\theta_c=\arcsin\frac{c_1}{c_{\rm S}}=38.6^\circ$.

6. For vertical incidence $\boldsymbol{k}'=\boldsymbol{0}$ and therefore $\xi_{{\rm I}n}=-\xi_{{\rm R}n}=\xi_{{\rm P}n}=\xi_{{\rm SV}n}=-1$. The reflection/transmission coefficients are solutions of:

$$\begin{gathered}1-R(\boldsymbol{0})=T_{\rm P}(\boldsymbol{0})+T_{\rm SV}(\boldsymbol{0})\,,\\ K_1k_1(1+R(\boldsymbol{0}))=\mu_2k_{\rm P}\kappa^2T_{\rm P}(\boldsymbol{0})\,,\\ \mu_2k_{\rm S}T_{\rm SV}(\boldsymbol{0})=0\,,\end{gathered}$$

   so that they are (with $\mathcal{Z}_{\rm P}=\varrho_2c_{\rm P}$):

$$R(\boldsymbol{0})=\frac{\mathcal{Z}_{\rm P}-\mathcal{Z}_1}{\mathcal{Z}_{\rm P}+\mathcal{Z}_1}\,,\quad T_{\rm P}(\boldsymbol{0})=\frac{2\mathcal{Z}_1}{\mathcal{Z}_{\rm P}+\mathcal{Z}_1}\,,\quad T_{\rm SV}(\boldsymbol{0})=0\,.$$

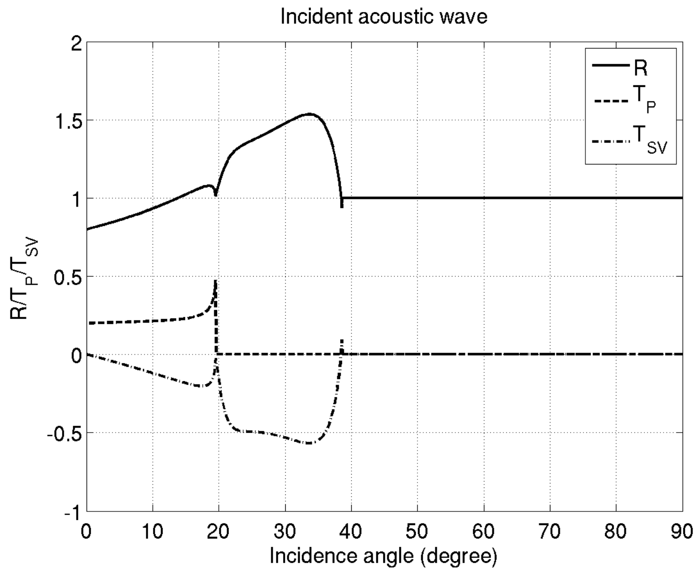

Figure 5: Reflection and transmission coefficients for an incident acoustic wave in a fluid medium over an elastic medium as functions of the incidence angle $\theta = \arcsin |\boldsymbol{\xi}'|$ for $\frac{\varrho_2}{\varrho_1} = \frac{c_P}{c_1} = 3$ and $\nu = 0.3$.